IB MATHEMATICS SL • UNIT 8.01

C Topic 8: IB SL Paper 1 & Paper 2 Extended-Response Synthesis

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

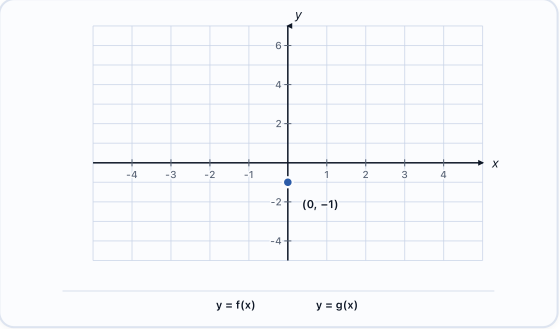
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 A	Q4 A	Q5 A
Q6 C	Q7 A	Q8 A	Q9 A	Q10 C
Q11 D	Q12 B	Q13 B	Q14 B	Q15 A
Q16 D	Q17 A	Q18 D	Q19 A	Q20 B
Q21 D	Q22 B	Q23 D	Q24 C	Q25 B

Topic 8: IB SL Paper 1 & Paper 2 Extended-Response Synthesis Practice Exercises

1 MCQ

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept?



A $(0, -3)$ ✓

B $(0, 3)$

C $(-1, 0)$

D $(3, 0)$

WORKED SOLUTION & VERIFICATION

The y-intercept of a function occurs when $x = 0$. Substituting $x = 0$ into the function $f(x) = -x^2 + 4x - 3$ gives $f(0) = -(0)^2 + 4(0) - 3 = -3$. Therefore, the coordinates of the y-intercept are $(0, -3)$. **[M1] [A1]**

2 MCQ

Consider the function $f(x) = 2x + 3$. What is the value of $f(4)$?

A 7

B 11 ✓

C 10

D 5

WORKED SOLUTION & VERIFICATION

To find the value of $f(4)$, we substitute $x = 4$ into the function $f(x) = 2x + 3$.

$f(4) = 2(4) + 3$

$f(4) = 8 + 3$

$f(4) = 11$ **[M1] [A1]**

3 MCQ

Consider the function $f(x) = 2x - 3$. Find the value of $f(3)$.

- **A** 3 ✓
- **B** 6
- **C** 9
- **D** 0

WORKED SOLUTION & VERIFICATION

To find the value of $f(3)$, we substitute $x = 3$ into the function definition $f(x) = 2x - 3$.

$f(3) = 2(3) - 3$
$f(3) = 6 - 3$
$f(3) = 3$. **[M1] [A1]**

4 MCQ

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.

- **A** **(3, 4)** ✓
- **B** (-3, 4)
- **C** (3, -4)
- **D** (4, 3)

WORKED SOLUTION & VERIFICATION

The given function is a quadratic of the form $f(x) = ax^2 + bx + c$, where $a = -1$, $b = 6$, and $c = -5$. The x-coordinate of the vertex of a parabola is given by $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{6}{2(-1)} = -\frac{6}{-2} = 3$. To find the y-coordinate of the vertex, substitute $x = 3$ into the function: $f(3) = -(3)^2 + 6(3) - 5 = -9 + 18 - 5 = 9 - 5 = 4$. Therefore, the coordinates of the vertex are (3, 4). **[M1] [A1]**

5 MCQ

Consider the function $f(x) = 2x - 1$. Find the value of $f(3)$.

- **A** 5 ✓
- **B** 6
- **C** 7
- **D** 4

WORKED SOLUTION & VERIFICATION

$f(x) = 2x - 1$. To find $f(3)$, substitute $x = 3$ into the function: $f(3) = 2(3) - 1$. $f(3) = 6 - 1$. $f(3) = 5$. Therefore, the value of $f(3)$ is 5. **[M1] [A1]**

6 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(0)$.

- **A** 1
- **B** -1
- **C** 3 ✓
- **D** -3

WORKED SOLUTION & VERIFICATION

To find the value of $f(0)$, substitute $x = 0$ into the function $f(x) = x^2 - 4x + 3$. $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$. Therefore, the value of $f(0)$ is 3. **[M1] [A1]**

7 MCQ

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the y-intercept.

- **A** 6 ✓
- **B** -3
- **C** 3
- **D** -6

WORKED SOLUTION & VERIFICATION

The y-intercept of a function is the value of y when $x = 0$. For the function $f(x) = -2x + 6$, we substitute $x = 0$ into the equation:

$f(0) = -2(0) + 6$

$f(0) = 0 + 6$

$f(0) = 6$

Therefore, the y-intercept is 6. **[M1] [A1]**

8 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

- **A** **(1, 0)** ✓
- **B** (3, 0)
- **C** (0, 3)
- **D** (0, 1)

WORKED SOLUTION & VERIFICATION

To find the x-intercepts of the function $f(x) = x^2 - 4x + 3$, we set $f(x) = 0$. This gives us the quadratic equation $x^2 - 4x + 3 = 0$. We can factor this equation as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. Therefore, the x-intercepts are at $(1, 0)$ and $(3, 0)$. The point with the smaller x-value is $(1, 0)$. **[M1] [A1]**

9 MCQ

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

A $y = 2x - 5$ ✓

B $y = -\frac{1}{2}x + 5$

C $y = 2x - 8$

D $y = -2x + 11$

WORKED SOLUTION & VERIFICATION

In a Voronoi diagram, the boundary edge between two sites is along the perpendicular bisector of the line segment connecting them.

1. Find the midpoint M of the segment AB:

$$M = (\frac{2+6}{2}, \frac{4+2}{2}) = (4, 3)$$

2. Find the gradient of segment AB:

$$m_{AB} = \frac{2-4}{6-2} = \frac{-2}{4} = -\frac{1}{2}$$

3. The gradient of the perpendicular bisector is the negative reciprocal of m_{AB}:

$$m_{\perp} = -\frac{1}{-\frac{1}{2}} = 2$$

4. Use the point-slope form with midpoint $(4, 3)$ and gradient $m_{\perp} = 2$:

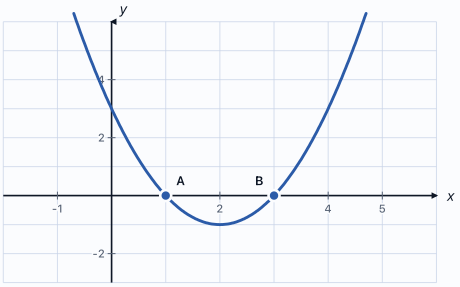
$$y - 3 = 2(x - 4) \implies y = 2x - 8 + 3 \implies y = 2x - 5$$

Common Misconceptions:

- Choice B uses the gradient of AB rather than its negative reciprocal.
- Choice C correctly finds the gradient $m = 2$ but forgets to add the y-coordinate of the midpoint, setting $y = 2(x - 4) = 2x - 8$.
- Choice D uses $m = -2$ instead of $+2$. **[M1] [A1]**

10 MCQ

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



A 1

B 3

C 2 ✓

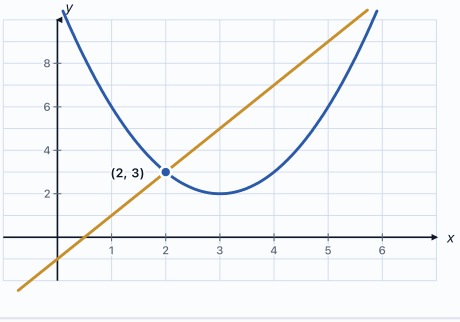
D 4

WORKED SOLUTION & VERIFICATION

The points where the graph intersects the x-axis are the roots of the quadratic equation $f(x) = 0$. We need to solve $x^2 - 4x + 3 = 0$. Factoring the quadratic, we get $(x - 1)(x - 3) = 0$. Thus, the roots are $x = 1$ and $x = 3$. Let point A have coordinates $(1, 0)$ and point B have coordinates $(3, 0)$. The distance between A and B is the absolute difference of their x-coordinates: $|3 - 1| = 2$. **[M1] [A1]**

11 MCQ

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.



A 3

B 4

C 5

D 6 ✓

WORKED SOLUTION & VERIFICATION

To find the intersection points of the two functions, we set $f(x) = g(x)$. $x^2 - 6x + 11 = 2x - 1$

Rearrange the equation to form a quadratic equation:

$x^2 - 6x - 2x + 11 + 1 = 0$

$x^2 - 8x + 12 = 0$

We can factor this quadratic equation. We are looking for two numbers that multiply to 12 and add to -8. These numbers are -6 and -2.

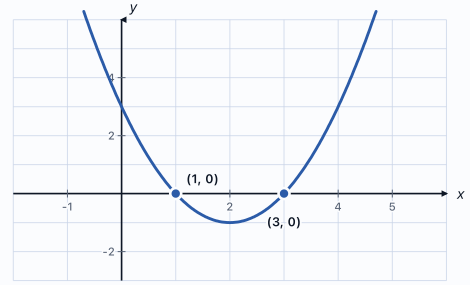
$(x - 2)(x - 6) = 0$

This gives us two possible solutions for x: $x = 2$ or $x = 6$.

We are given that one intersection point has an x-coordinate of 2. Therefore, the x-coordinate of the other intersection point is 6. **[M1] [A1]**

12 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. Find the sum of the x-intercepts of the graph of $y = f(x)$.



A 3

B 4 ✓

C -4

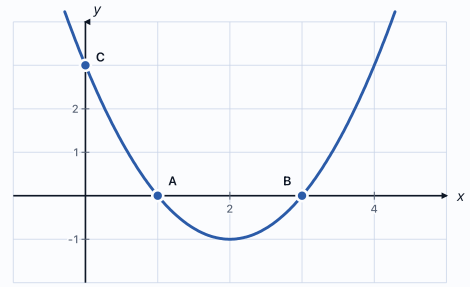
D 1

WORKED SOLUTION & VERIFICATION

The x-intercepts of the graph of $y = f(x)$ occur when $f(x) = 0$. So, we need to solve the quadratic equation $x^2 - 4x + 3 = 0$. This equation can be factored as $(x - 1)(x - 3) = 0$, which gives the x-intercepts as $x = 1$ and $x = 3$. The sum of the x-intercepts is $1 + 3 = 4$. Alternatively, for a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this case, $a = 1$, $b = -4$, and $c = 3$. Thus, the sum of the roots is $-\frac{-4}{1} = 4$. **[M1] [A1]**

13 MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the sum of the x-coordinates of A and B.



A -4

B 4 ✓

C 3

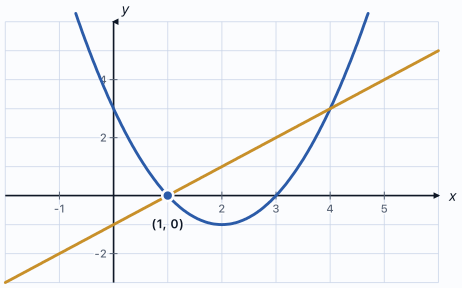
D 1

WORKED SOLUTION & VERIFICATION

For the graph of a quadratic function $f(x) = ax^2 + bx + c$, the sum of the roots (x-intercepts) is given by $-\frac{b}{a}$. In this case, $f(x) = x^2 - 4x + 3$, so $a = 1$ and $b = -4$. Therefore, the sum of the x-coordinates of the points where the graph intersects the x-axis is $-(\frac{-4}{1}) = 4$. Alternatively, we can find the roots by setting $f(x) = 0$: $x^2 - 4x + 3 = 0$. Factoring the quadratic gives $(x - 1)(x - 3) = 0$. The roots are $x = 1$ and $x = 3$. The sum of the roots is $1 + 3 = 4$. **[M1] [A1]**

14 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(5)$.



A 1

B 8 ✓

C 0

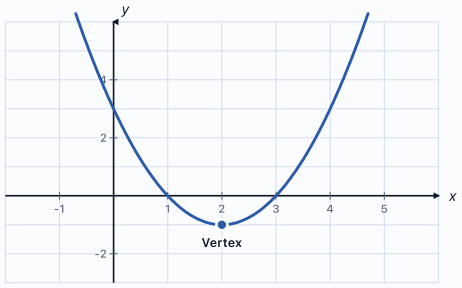
D 18

WORKED SOLUTION & VERIFICATION

To find the value of $f(5)$, we substitute $x = 5$ into the function $f(x) = x^2 - 4x + 3$.
$f(5) = (5)^2 - 4(5) + 3$
$f(5) = 25 - 20 + 3$
$f(5) = 5 + 3$
$f(5) = 8$
Therefore, the value of $f(5)$ is 8. Common errors might arise from miscalculating the square, the subtraction, or the addition. **[M1] [A1]**

15 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.



A 2 ✓

B -2

C 1

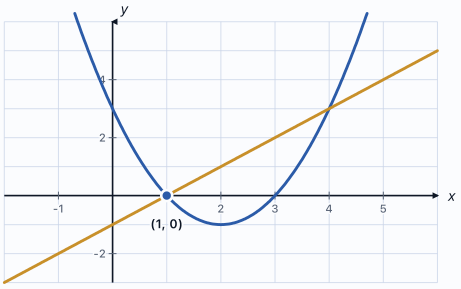
D 3

WORKED SOLUTION & VERIFICATION

The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -b/(2a)$. In this case, $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -(-4)/(2 * 1) = 4/2 = 2$. **[M1] [A1]**

16 MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.



A (3, 2)

B (5, 4)

C (2, 1)

D **(4, 3)** ✓

WORKED SOLUTION & VERIFICATION

To find the intersection points of the two graphs, we set $f(x) = g(x)$.

$x^2 - 4x + 3 = x - 1$

Subtract x from both sides:

$x^2 - 5x + 3 = -1$

Add 1 to both sides:

$x^2 - 5x + 4 = 0$

Factor the quadratic equation:

$(x - 1)(x - 4) = 0$

This gives us two possible x-coordinates for the intersection points: $x = 1$ and $x = 4$.

We are given that one intersection point has an x-coordinate of 1. To find the corresponding y-coordinate, we can substitute $x = 1$ into either $f(x)$ or $g(x)$.

Using $g(x) = x - 1$, we get $g(1) = 1 - 1 = 0$. So the first intersection point is $(1, 0)$.

To find the other intersection point, we use the other x-coordinate, $x = 4$.

Substituting $x = 4$ into $g(x) = x - 1$, we get $g(4) = 4 - 1 = 3$.

Therefore, the other intersection point is $(4, 3)$. **[M1] [A1]**

17 MCQ

$\$1200$ is invested at a simple interest rate of 8% per year for 5 years. Find the interest earned.

A **480.00** ✓

B 470.00

C 490.00

D 500.00

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $I = \frac{Prt}{100} = \frac{(1200)(8)(5)}{100} = 480.00$. **[M1] [A1]**

18 MCQ

Find the midpoint of $A(-1,-1)$ and $B(-7,5)$.

A $(-6,6)$

B $(-3,2)$

C $(-4,3)$

D $(-4,2)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M=(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2})=(\frac{-1-7}{2},\frac{-1+5}{2})=(-4,2)$. **[M1] [A1]**

19 MCQ

Find the midpoint of $A(1,2)$ and $B(5,10)$.

A $(3,6)$ ✓

B $(3,7)$

C $(4,8)$

D $(4,6)$

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M=(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2})=(\frac{1+5}{2},\frac{2+10}{2})=(3,6)$. **[M1] [A1]**

20 MCQ

$\$500$ is invested at a simple interest rate of 6% per year for 5 years. Find the interest earned.

A 160.00

B **150.00** ✓

C 170.00

D 140.00

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $I=\frac{Prt}{100}=\frac{(500)(6)(5)}{100}=150.00$. **[M1] [A1]**

21 MCQ

Calculate the midpoint of $A(8,7)$ and $B(2,13)$.

A $(-6,6)$

B $(6,10)$

C $(5,11)$

D $(5,10)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M=(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2})=(\frac{8+2}{2},\frac{7+13}{2})=(5,10)$. **[M1] [A1]**

22 MCQ

Find the midpoint of $A(1, -6)$ and $B(-9, 2)$.

A $(-4, -1)$

B $(-4, -2)$ ✓

C $(-10, 8)$

D $(-3, -2)$

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}) = (\frac{1 - 9}{2}, \frac{-6 + 2}{2}) = (-4, -2)$. **[M1] [A1]**

23 MCQ

Find the midpoint of $A(-3, -8)$ and $B(-7, -10)$.

A $(-4, -9)$

B $(-4, -2)$

C $(-5, -8)$

D $(-5, -9)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}) = (\frac{-3 - 7}{2}, \frac{-8 - 10}{2}) = (-5, -9)$. **[M1] [A1]**

24 MCQ

$\$2000$ is invested at a simple interest rate of 2% per year for 3 years. Find the interest earned.

A 110.00

B 140.00

C **120.00** ✓

D 130.00

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $I = \frac{Prt}{100} = \frac{(2000)(2)(3)}{100} = 120.00$. **[M1] [A1]**

25 MCQ

$\$500$ is invested at a simple interest rate of 4% per year for 2 years. Find the interest earned.

A 30.00

B **40.00** ✓

C 50.00

D 60.00

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $I = \frac{Prt}{100} = \frac{(500)(4)(2)}{100} = 40.00$. **[M1] [A1]**